SOLUTIONS: SYSTEMS IDENTIFICATION AND LEARNING

1. Solution

a) The transfer function associated with the difference equation $v(t) = -\frac{1}{3}v(t-1) + e(t) + ce(t-1)$ is given by

$$V(z) = \frac{1+cz^{-1}}{1+\frac{1}{3}z^{-1}}E(z) = \frac{z+c}{z+1/3}E(z)$$

Since the pole p of the transfer function $\dfrac{z+c}{z+1/3}$ is $p = -1/3$, then $|p| = 1/3 < 1$ and hence this transfer function represents an asymptotically stable system. Therefore, the stochastic process $v(\cdot)$ is stationary.

[3 Marks]

b) We have:

$$\begin{aligned}
\gamma(0) &= \text{var}[v(t)] \\
\gamma(1) &= \mathbb{E}[v(t)v(t-1)] = \mathbb{E}\left\{\left[-\tfrac{1}{3}v(t-1) + e(t) + ce(t-1)\right]\cdot v(t-1)\right\} \\
&= -\tfrac{1}{3}\mathbb{E}[v(t-1)^2] + c\mathbb{E}[v(t-1)e(t-1)] \\
&= -\tfrac{1}{3}\gamma(0) + c
\end{aligned} \tag{1.1}$$

Thus

$$\rho(1) = -\frac{1}{3} \Longrightarrow \frac{\gamma(1)}{\gamma(0)} = -\frac{1}{3} \Longrightarrow c = 0$$

[3 Marks]

c) Imposing $c = 0$ into $v(t) = -\frac{1}{3}v(t-1) + e(t) + ce(t-1)$ gives $v(t) = -\frac{1}{3}v(t-1) + e(t)$ and hence

$$A(z)v(t) = C(z)e(t) \quad \text{with} \quad A(z) := 1 + \frac{1}{3}z^{-1}; C(z) := 1$$

By performing two steps of polynomial division of $C(z)$ by $A(z)$ we have

$$\widehat{W}(z) := \frac{C(z)}{A(z)} = 1 - \frac{1}{3}z^{-1} + z^{-2}\frac{\frac{1}{9}}{1+\frac{1}{3}z^{-1}}$$

thus obtaining

$$\hat{v}(t+2|t) = \frac{1}{9}v(t)$$

Answers

[5 Marks]

d) From (1.1) given in your answer to Question 1b) and taking into account that now $\rho(1) = 3/10$, we have

$$\gamma(1) = -\frac{1}{3}\gamma(0) + c \Longrightarrow -\frac{1}{3}\gamma(0) + c = \frac{3}{10}\gamma(0) \Longrightarrow c = \frac{19}{30}\gamma(0)$$

Now:

$$\begin{aligned} \gamma(0) \quad &= \mathbb{E}\left\{\left[-\tfrac{1}{3}v(t-1) + e(t) + ce(t-1)\right]^2\right\} \\ &= \tfrac{1}{9}\gamma(0) + 1 + c^2 - \tfrac{2}{3}c \Longrightarrow \gamma(0) = \tfrac{9}{8}(1 + c^2 - \tfrac{2}{3}c) \end{aligned} \tag{1.2}$$

and replacing $c = \frac{19}{30}\gamma(0)$ into (1.2) above, we obtain the polynomial equation

$$\frac{361}{800}\gamma(0)^2 - \frac{59}{40}\gamma(0) + \frac{9}{8} = 0 \tag{1.3}$$

Equation (1.3) has two solutions:

$$\gamma(0)_{(1)} = \frac{10}{361}(59 - 2\sqrt{58}); \quad \gamma(0)_{(2)} = \frac{10}{361}(59 + 2\sqrt{58})$$

yielding two possible values for the parameter c:

$$c_{(1)} = \frac{1}{57}(59 - 2\sqrt{58}) \simeq 0.768; \quad c_{(2)} = \frac{1}{57}(59 + 2\sqrt{58}) \simeq 1.3;$$

As $c_{(1)} < 1$ and $c_{(2)} > 1$, only $c_{(1)} < 1$ is an admissible value for the parameter c because the ARMA model has to be in spectral canonical form.

[6 Marks]

e) The ARMA model now is:

$$A(z)v(t) = C(z)e(t) \quad \text{with} \quad A(z) := 1 + \frac{1}{3}z^{-1}; C(z) := 1 + \frac{1}{57}(59 - 2\sqrt{58})z^{-1}$$

The one-step ahead optimal prediction $\hat{v}(t+1|t)$ can be obtained by using the general formula

$$\hat{v}(t+1|t) = \frac{C(z) - A(z)}{C(z)}v(t+1) = \frac{\frac{1}{57}(59 - 2\sqrt{58}) - 1/3}{1 + \frac{1}{57}(59 - 2\sqrt{58})z^{-1}}z^{-1}v(t+1)$$

yielding

$$\hat{v}(t+1|t) = -\frac{1}{57}(59 - 2\sqrt{58})\hat{v}(t|t-1) + \left[\frac{1}{57}(59 - 2\sqrt{58}) - 1/3\right]v(t)$$

[3 Marks]

Answers

2. Solution

a) By inspecting the block-scheme shown in Fig. 2.1 in the text of Question 2 of the exam paper, one immediately gets:

$$Y(z) = 3\left[\frac{1}{3}\cdot\left(1-\frac{9}{10}z^{-1}\right)V(z)+\frac{1}{3}\cdot\frac{1}{1+\frac{1}{2}z^{-1}}W(z)\right]$$

Simple algebra gives

$$Y(z) = \frac{\left(1-\frac{9}{10}z^{-1}\right)\cdot\left(1+\frac{1}{2}z^{-1}\right)}{1+\frac{1}{2}z^{-1}}\cdot V(z)+\frac{1}{1+\frac{1}{2}z^{-1}}W(z)$$

and thus

$$Y(z) = \frac{1-\frac{2}{5}z^{-1}-\frac{9}{20}z^{-2}}{1+\frac{1}{2}z^{-1}}\cdot V(z)+\frac{1}{1+\frac{1}{2}z^{-1}}W(z) \qquad (2.1)$$

Therefore, one gets:

$$y(t) = -\frac{1}{2}y(t-1)+w(t)+v(t)-\frac{2}{5}v(t-1)-\frac{9}{20}v(t-2) \qquad (2.2)$$

[4 Marks]

b) Since the pole of the two transfer functions in Equation (2.1) in the answer to Question 2a) lies strictly inside the unit circle, it follows that the stochastic process $y(\cdot)$ given by the model shown by Equation (2.2) in the answer to Question 2a) is stationary.

By applying the expected value operator on both sides of Equation (2.2) in the answer to Question 2a), we obtain

$$\mathbb{E}[y(t)] = -\frac{1}{2}\mathbb{E}[y(t-1)]+\mathbb{E}[w(t)]+\mathbb{E}[v(t)]-\frac{2}{5}\mathbb{E}[v(t-1)]-\frac{9}{20}\mathbb{E}[v(t-2)]$$

and, due to the stationarity of the process $y(\cdot)$, we can easily conclude that $\mathbb{E}[y(t)] = 0$.

[4 Marks]

c) Since we have shown that $\mathbb{E}[y(t)] = 0$, again due to the stationarity of the process generated by (2.2), and by exploiting the whiteness and the mutual uncorrelation of the processes $v(\cdot)$ and $w(\cdot)$, we obtain

$$\begin{aligned}\text{var}[y(t)] &= \mathbb{E}[y(t)^2] = \mathbb{E}\left\{\left[-\frac{1}{2}y(t-1)+w(t)+v(t)-\frac{2}{5}v(t-1)-\frac{9}{20}v(t-2)\right]^2\right\}\\ &= \frac{1}{4}\text{var}[y(t)]+\frac{1033}{400} \Longrightarrow \text{var}[y(t)] = \frac{1033}{300} \simeq 3.44\end{aligned}$$

[4 Marks]

Answers

d) Since

$$Y(z) = \left(1 - \frac{9}{10}z^{-1}\right)V(z) + \frac{1}{1+\frac{1}{2}z^{-1}}W(z)$$

and owing to the fact that $v(\cdot) \sim WN(0,1)$, $w(\cdot) \sim WN(0,1)$, and that $e(\cdot)$ and $w(\cdot)$ are uncorrelated, it follows that

$$\Gamma_y(\omega) = \left[\frac{1}{(1+\frac{1}{2}z^{-1})(1+\frac{1}{2}z)} + \left(1 - \frac{9}{10}z^{-1}\right)\left(1 - \frac{9}{10}z\right)\right]_{z=e^{j\omega}}$$

which, after some algebra, gives

$$\Gamma_y(\omega) = \frac{181}{100} - \frac{9}{5}\cos\omega + \frac{1}{\frac{5}{4}+\cos\omega}$$

As

$$Y(z) = \left(1 - \frac{9}{10}z^{-1}\right)V(z) + \frac{1}{1+\frac{1}{2}z^{-1}}W(z) = \frac{z-\frac{9}{10}}{z}V(z) + \frac{z}{z+\frac{1}{2}}W(z) +$$

we obtain

$$\Gamma_y(\omega) = \frac{1}{\left|e^{j\omega}+\frac{1}{2}\right|^2} + \left|e^{j\omega} - \frac{9}{10}\right|^2$$

Thus, a few values taken on by the spectrum can be easily computed by geometric considerations:

$$\Gamma_y(0) = \frac{1}{(1+\frac{1}{2})^2} + \frac{1}{100} = \frac{409}{900} \simeq 0.45$$

$$\Gamma_y(\pi/2) = \frac{1}{1+\frac{1}{4}} + 1 + \frac{81}{100} = \frac{261}{100} \simeq 2.61$$

$$\Gamma_y(\pi) = \frac{1}{\frac{1}{4}} + \left(1 + \frac{9}{10}\right)^2 = \frac{761}{100} \simeq 7.61$$

Answers

The spectrum behaviour in the interval $\omega \in [-\pi, \pi]$ is given in Fig. 2.1

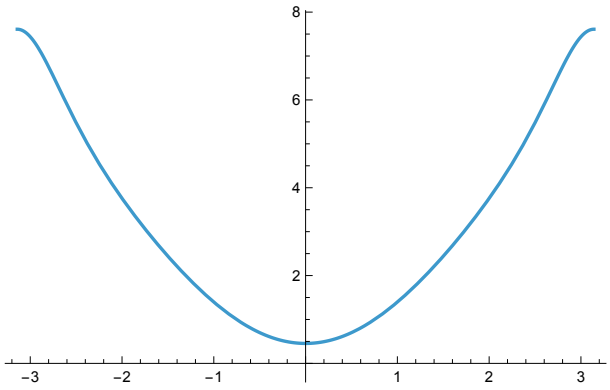


Figure 2.1 Plot of the spectrum $\Gamma_y(\omega) = \frac{181}{100} - \frac{9}{5}\cos\omega + \frac{1}{\frac{5}{4}+\cos\omega}$.

[8 Marks]

Answers

3. Solution

a) One refers to Case 1 in the text of the exam paper where $y(\cdot) = v_1(\cdot)$. The model in prediction form is

$$\widehat{\mathscr{M}}(\theta): \quad \hat{y}(t|t-1) = ay(t-1)$$

The estimate $\hat{a}(N)$ converges almost surely to the minima of

$$\bar{J}(a) \quad = \mathbb{E}\left\{[y(t)-\hat{y}(t|t-1)]^2\right\} = \mathbb{E}\left\{[y(t)-ay(t-1)]^2\right\}$$

$$= \mathbb{E}[y(t)^2] - 2a\mathbb{E}[y(t)y(t-1)] + a^2\mathbb{E}[y(t-1)^2] = (1+a^2)\gamma_y(0) - 2a\gamma_y(1)$$

where $\gamma_y(\tau)$ denotes the correlation function of the process $y(\cdot)$.

Therefore, $\bar{J}(a)$ has a single minimum attained for $\bar{a} = \gamma_y(1)/\gamma_y(0)$. One computes $\gamma_y(1)$ as follows:

$$\gamma_y(1) = \mathbb{E}[y(t)y(t-1)] = \mathbb{E}\left\{\left[-\frac{1}{3}y(t-1)+w(t)\right]y(t-1)\right\} = -\frac{1}{3}\gamma_y(0)$$

Thus $\bar{a}_1 = \gamma_y(1)/\gamma_y(0) = -\frac{1}{3}$.

[6 Marks]

b) Now, one refers to Case 2 in the text of the exam paper, where $y(\cdot) = v_2(\cdot)$, and computes $\gamma_y(0)$ and $\gamma_y(1)$:

$$\gamma_y(0) \quad = \mathbb{E}[y(t)^2] = \mathbb{E}\left\{\left[-\tfrac{1}{3}y(t-1)+w(t)-\tfrac{1}{2}w(t-1)\right]^2\right\}$$

$$= \tfrac{1}{9}\gamma_y(0) + \mathrm{var}[w(t)] + \tfrac{1}{4}\mathrm{var}[\xi(t)] + \tfrac{1}{3}\mathbb{E}[y(t-1)w(t-1)]$$

Hence $\gamma_y(0) = \frac{57}{32}\mathrm{var}[\xi(t)] = \frac{57}{32}$.

$$\gamma_y(1) \quad = \mathbb{E}[y(t)y(t-1)] = \mathbb{E}\left\{\left[-\tfrac{1}{3}y(t-1)+w(t)-\tfrac{1}{2}w(t-1)\right]y(t-1)\right\}$$

$$= -\tfrac{1}{3}\gamma_y(0) - \tfrac{1}{2}\mathrm{var}[w(t)] = -\tfrac{3}{32}\mathrm{var}[w(t)]$$

Thus $\bar{a}_2 = \gamma_y(1)/\gamma_y(0) = -\frac{1}{19}\mathrm{var}[w(t)] = -\frac{1}{19}$.

[7 Marks]

c) We have:

$$\mathrm{var}[v_1(t) - \bar{a}_1 v_1(t-1)] = \mathbb{E}\left\{\left[-\frac{1}{3}v_1(t-1)+w(t)+\frac{1}{3}v_1(t-1)\right]^2\right\} = \mathrm{var}[w(t)]$$

$$\mathrm{var}[v_2(t) - \bar{a}_2 v_2(t-1)] \quad = \mathbb{E}\left\{\left[-\tfrac{22}{57}v_2(t-1)+w(t)-\tfrac{1}{2}w(t-1)\right]^2\right\}$$

$$= \tfrac{23197}{12996}\mathrm{var}[\xi(t)] \simeq 1.78 \cdot \mathrm{var}[w(t)]$$

Answers

As should be expected because of the presence of the coloured noise in model generating $v_2(\cdot)$, the prediction error $v_2(t) - \bar{a}_2 v_2(t-1)$ is not white and

$$\text{var}[\varepsilon_2(t)] = \frac{23197}{12996}\text{var}[w(t)] \simeq 1.78 > 1 = \text{var}[w(t)] = \text{var}[\varepsilon_1(t)]$$

[4 Marks]

Let us address the case $w(\cdot) \sim WN(0,3)$. As we have seen in the previous answers, $\gamma_y(0)$ and $\gamma_y(1)$ are always proportional to $\text{var}[w(t)]$ and hence the values $\bar{a}_1 = \gamma_y(1)/\gamma_y(0)$ when $y(\cdot) = v_1(\cdot)$ and $\bar{a}_2 = \gamma_y(1)/\gamma_y(0)$ when $y(\cdot) = v_2(\cdot)$ do not depend on $\text{var}[w(t)]$.

[3 Marks]

Answers

4. a) i) According to the text of Question 4-a), the parametric model $\mathscr{M}_1$ can be written as

$$\mathscr{M}_1 : \widehat{c}[V(j),\theta] = \Psi[V(j)]\cdot\theta,\ j=1,\ldots,M$$

where $\Psi[V(j)]=1$, $j=1,\ldots,M$ and $\theta=\alpha$.

Introduce the error variable given by

$$e(j) = c_m(j) - \widehat{c}[V(j),\theta] = c_m(j) - \Psi[t(j)]\theta\,,\ j=1,\ldots,M\,.$$

Moreover, consider the following cost function:

$$J(\theta) = \frac{1}{M}\sum_{j=1}^{M}[e(j)]^2 = \frac{1}{M}\sum_{j=1}^{M}\{c_m(j)-\Psi[t(j)]\theta\}^2\,.$$

The minimisation of $J(\theta)$ with respect to the unknown scalar θ yields the optimal (in the least-squares sense) parameter θ of model $\mathscr{M}_1$ provided that a unique minimum of $J(\theta)$ does exist.

To compute the optimal solution θ°, the gradient of $J(\theta)$ with respect to the scalar θ is determined:

$$\begin{aligned}\frac{\partial}{\partial\alpha}J(\theta) = \frac{\partial}{\partial\alpha}J(\alpha) &= \frac{1}{M}\sum_{j=1}^{M}\frac{\partial}{\partial\alpha}\left\{[c_m(j)-\Psi[V(j)]\alpha]^2\right\} = \\ &= \frac{2}{M}\sum_{j=1}^{M}[c_m(j)-\Psi[V(j)]\alpha]\frac{\partial}{\partial\alpha}[c_m(j)-\Psi[V(j)]\alpha] = \\ &= -\frac{2}{M}\sum_{j=1}^{M}[c_m(j)-\Psi[V(j)]\alpha]\Psi[V(j)]\end{aligned}$$

Now, by imposing

$$\frac{\partial}{\partial\alpha}J(\alpha)=0\,,$$

after some algebra, the following linear problem is obtained:

$$M\cdot\alpha = \sum_{j=1}^{M}c_m(j)$$

that has the unique solution

$$\alpha^\circ = \frac{1}{M}\sum_{j=1}^{M}c_m(j)$$

[5 Marks]

ii) The parametric model $\mathscr{M}_2$ can be written as

$$\mathscr{M}_2 : \widehat{c}[V(j),\theta] = \Psi[V(j)]^\top\cdot\theta,\ j=1,\ldots,M$$

where

$$\Psi[V(j)] := \begin{bmatrix}1\\ \arctan[V(j)]\end{bmatrix};\quad \theta := \begin{bmatrix}\alpha\\ \beta\end{bmatrix}$$

Answers

Introduce the error variable given by

$$e(j) = c_m(j) - \widehat{c}[V(j),\theta] = c_m(j) - \Psi[V(j)]^\top \cdot \theta\,,\ j = 1,\ldots,M\,.$$

Moreover, consider the following cost function:

$$J(\theta) = \frac{1}{M}\sum_{j=1}^{M}[e(j)]^2 = \frac{1}{M}\sum_{j=1}^{M}\{c_m(j) - \Psi[V(j)]^\top \cdot \theta\}^2\,.$$

The minimisation of $J(\theta)$ with respect to the unknown vector θ yields the optimal (in the least-squares sense) set of parameters α and β of model $\mathscr{M}_2$ provided that a unique minimum of $J(\theta)$ does exist.

To compute the optimal solution $\theta^\circ = [\alpha^\circ, \beta^\circ]^\top$, the gradient of $J(\theta)$ with respect to the vector θ is determined:

$$\begin{aligned}\frac{\partial}{\partial\theta}J(\theta) &= \frac{1}{M}\sum_{j=1}^{M}\frac{\partial}{\partial\theta}\left\{[c_m(j) - \Psi[V(j)]^\top \cdot \theta]^2\right\} = \\ &= \frac{2}{M}\sum_{j=1}^{M}[c_m(j) - \Psi[V(j)]^\top \cdot \theta]\frac{\partial}{\partial\theta}\left\{c_m(j) - \Psi[V(j)]^\top \cdot \theta\right\} = \\ &= -\frac{2}{M}\sum_{j=1}^{M}[c_m(j) - \Psi[V(j)]^\top \cdot \theta]\Psi[V(j)]^\top\end{aligned}$$

Now, by imposing

$$\frac{\partial}{\partial\theta}J(\theta) = [0\,,0]\,,$$

after some algebra, the following linear problem is obtained:

$$\left\{\sum_{j=1}^{M}\Psi[V(j)]\Psi[V(j)]^\top\right\}\theta = \sum_{j=1}^{M}c_m(j)\Psi[V(j)]$$

that has a unique solution θ° if the matrix

$$\sum_{j=1}^{M}\Psi[V(j)]\Psi[V(j)]^\top$$

is non-singular.

Finally, using the specific form of vector $\Psi[V(j)]$, one has:

$$\sum_{j=1}^{M}\Psi[t(j)]\Psi[t(j)]^\top = \begin{bmatrix} M & \displaystyle\sum_{j=1}^{M}\arctan[V(j)] \\ \displaystyle\sum_{j=1}^{M}\arctan[V(j)] & \displaystyle\sum_{j=1}^{M}\{\arctan[V(j)]\}^2 \end{bmatrix}$$

and

$$\sum_{j=1}^{M}c_m(j)\Psi[V(j)] = \begin{bmatrix} \displaystyle\sum_{j=1}^{M}c_m(j) \\ \displaystyle\sum_{j=1}^{M}c_m(j)\cdot\arctan[V(j)] \end{bmatrix}$$

[7 Marks]

Answers

b) Consider the set Θ given in the text of Question 4-b) of the exam paper:

$$\Theta = \{(1,1.34),(2,1.58),(3,1.75),(4,1.79),(5,1.9)\}$$

and substitute the numerical values of $V(j)$ and $c_m(j)$ into the general solutions obtained in the answers to Question 4-a-i) and Question 4-a-ii), respectively.

i)

$$\alpha^\circ = \frac{1}{5}\sum_{j=1}^{5} c_m(j) \simeq 1.672$$

and hence the optimal (in the least-squares sense) model $\mathscr{M}_1$ is given by:

$$\mathscr{M}_1 : \widehat{c}[V(j),\alpha] \simeq 1.672,\ j = 1,\ldots,5 \text{ (Constant Model)}$$

Fig. 4.1 shows the diagram of model $\mathscr{M}_1$ (cyan horizontal line) and the five noisy measurements c_m of the current belonging to the set Θ. It can be noticed that the approximate model $\mathscr{M}_1$ fits badly the measurements because of the poor representation capabilities of a constant model (basically kind of neglecting the expected/typical physical behaviour of the current for increasing values of the voltage). In Fig. 4.1, the time-behaviour of the true current is also shown (blue continuous line) for completeness and for the sake of comparison.

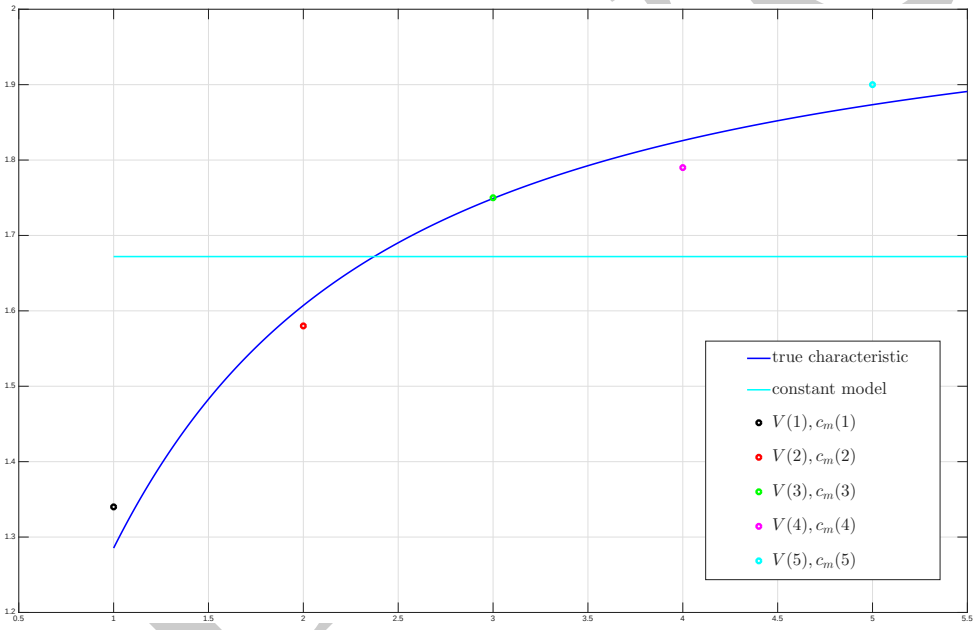


Figure 4.1

Remark for the students (not part of the solution of the exam paper).

In Fig. 4.1, the time-behaviour of the true current is also shown (blue continuous line) for completeness and for the sake of comparison.

[3 Marks]

ii) Using again the set Θ given in the text of Question 4-b) of the exam paper:

$$\Theta = \{(1,1.34),(2,1.58),(3,1.75),(4,1.79),(5,1.9)\}$$

after some easy calculations we get:

$$\sum_{j=1}^{M}\Psi[t(j)]\Psi[t(j)]^{\top}=\begin{bmatrix} M & \sum_{j=1}^{M}\arctan[V(j)] \\ \sum_{j=1}^{M}\arctan[V(j)] & \sum_{j=1}^{M}\{\arctan[V(j)]\}^2 \end{bmatrix}\simeq\begin{bmatrix} 5 & 5.84 \\ 5.84 & 7.05 \end{bmatrix}$$

and

$$\sum_{j=1}^{M}c_m(j)\Psi[V(j)]=\begin{bmatrix} \sum_{j=1}^{M}c_m(j) \\ \sum_{j=1}^{M}c_m(j)\cdot\arctan[V(j)] \end{bmatrix}=\begin{bmatrix} 8.36 \\ 9.97 \end{bmatrix}$$

and hence the optimal (in the least-squares sense) model $\mathscr{M}_2$ is given by:

$\mathscr{M}_2:\ \widehat{c}[V(j),\alpha]\simeq 0.6+0.91\cdot\arctan[V(j)],\ j=1,\ldots,5$ (Nonlinear Model)

Fig. 4.2 shows the diagram of model $\mathscr{M}_2$ (cyan line) and the five noisy measurements c_m of the current belonging to the set Θ. It can be noticed that the approximate model $\mathscr{M}_2$ fits very well the measurements because of the good representation capabilities of a nonlinear model. In Fig. 4.2, the time-behaviour of the true current is also shown (blue continuous line) for completeness and for the sake of comparison.

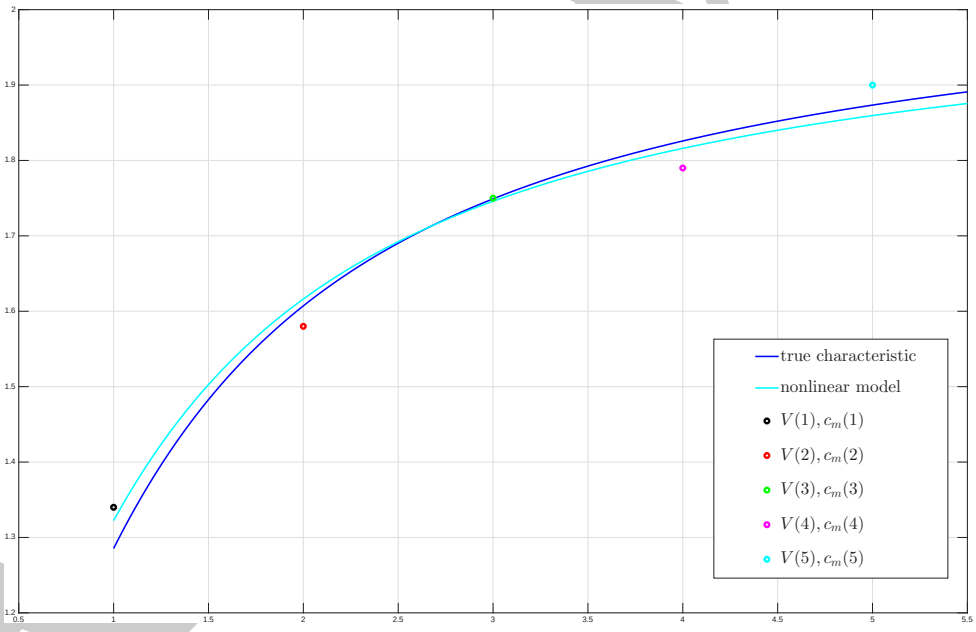


Figure 4.2

Remark for the students (not part of the solution of the exam paper).

In Fig. 4.2, the time-behaviour of the true current is also shown (continuous line) for completeness and for the sake of comparison.

[5 Marks]

Answers